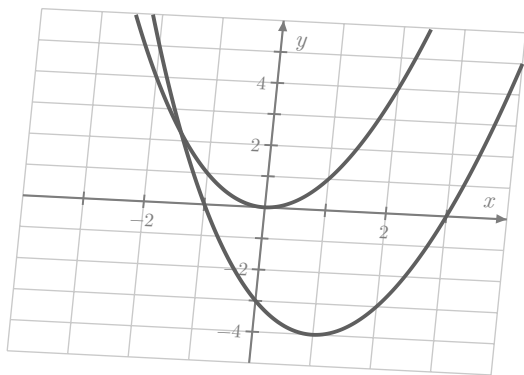
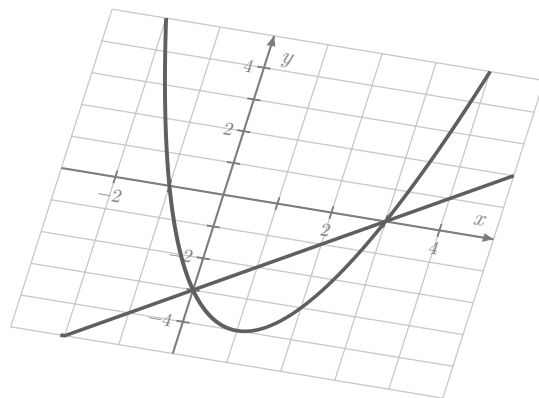




$$f(x) = a(x-h)^2 + k$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



$$x = -\frac{b}{2a}$$

$$\begin{aligned} x^2 - 2x - 3 &= 0 \\ (x+1)(x-3) &= 0 \\ x &= -1 \quad x = 3 \end{aligned}$$

$$x^2 = k \Rightarrow x = \pm \sqrt{k}$$

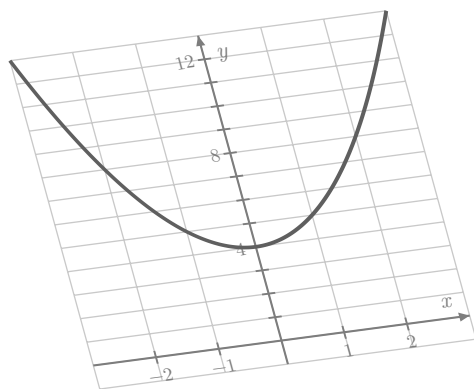
$$f(x) = a(x-p)(x-q)$$

$$\begin{aligned} x^2 - 2x &= 3 \\ (x-1)^2 &= 4 \\ x &= 1 \pm 2 \end{aligned}$$

Algebra 2

Unit 3: Quadratic Functions

Shepherd



$$i = \sqrt{-1} \quad i^2 = -1$$

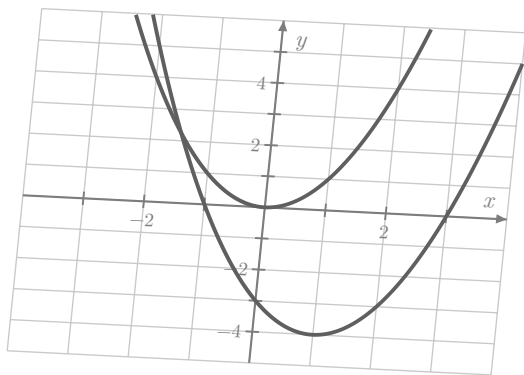
$$(a+bi)(a-bi) = a^2 + b^2$$

$$\begin{aligned} x^2 + 2x + 5 &= 0 \\ (x+1)^2 &= -4 \\ x &= -1 \pm 2i \end{aligned}$$

$$h(t) = -16t^2 + 32t + 48$$

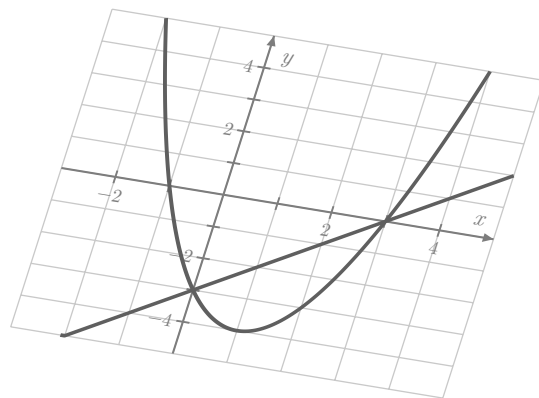
$$A(x) = x(40 - 2x)$$

$$\begin{aligned} b^2 - 4ac &> 0 \\ b^2 - 4ac &= 0 \\ b^2 - 4ac &< 0 \end{aligned}$$



$$f(x) = a(x-h)^2 + k$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



$$x = -\frac{b}{2a}$$

$$\begin{aligned} x^2 - 2x - 3 &= 0 \\ (x+1)(x-3) &= 0 \\ x &= -1 \quad x = 3 \end{aligned}$$

$$x^2 = k \Rightarrow x = \pm \sqrt{k}$$

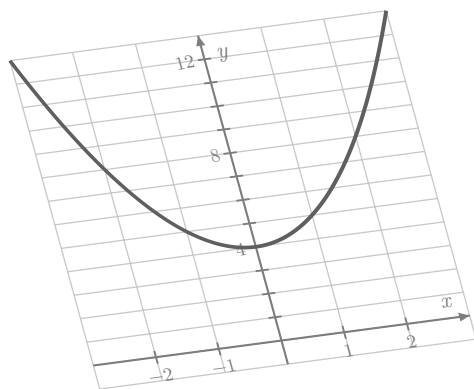
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Algebra 2

Unit 3: Quadratic Functions

Shepherd



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